

Regresión Lineal (Simple)

Optimización

Modelo

(Predicción $\hat{y}$)

$$\hat{y} = \beta_0 + \beta_1 x$$

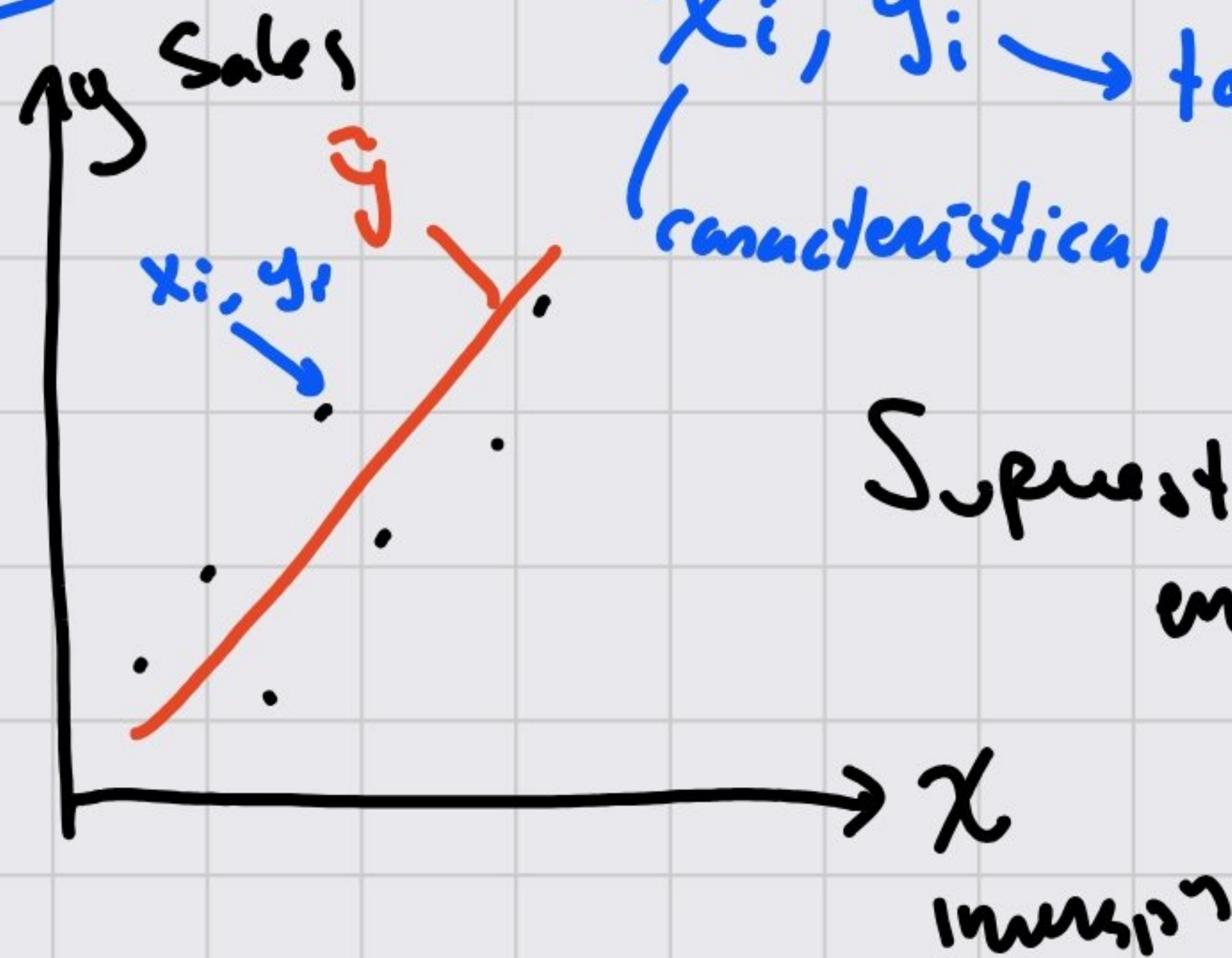
$$\hat{y} = \log(x)$$

$$\hat{y} = e^x$$

$$\hat{y} = x^p$$

DATOS

$x_i, y_i \rightarrow$ target
(características)



Supuesto, hay corr. lineal
entre x y y .

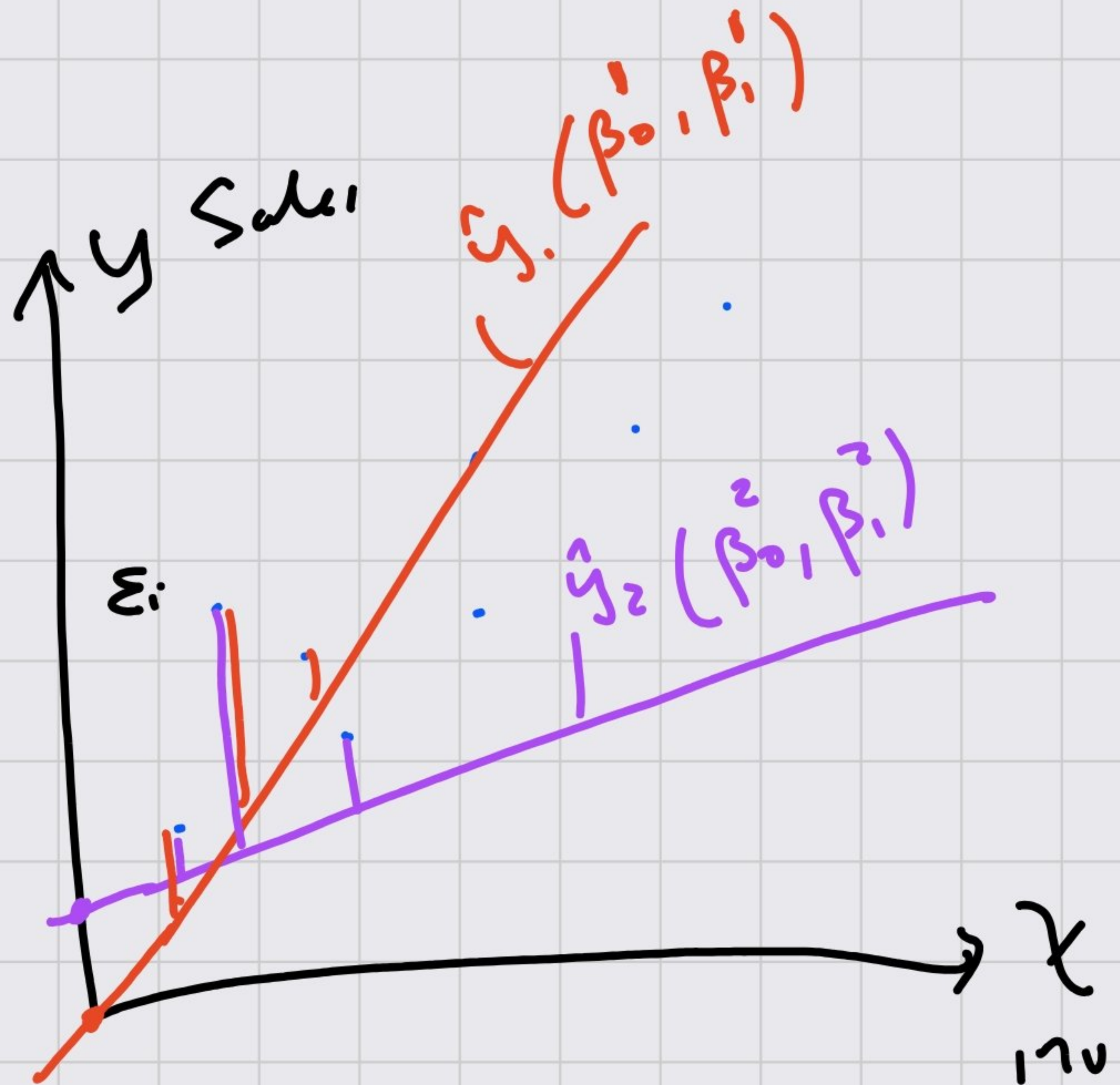
$$\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$$

Pearson

[1, 1]

$$\hat{y} = \beta_0 + \beta_1 x$$

$+ \beta_1$
 $+ 1$
 Sales
 Parametros
 Ajustar.
 Inversion $n \times n$
 $\beta_0 \rightarrow$ ordenada al origen
Intercepto
 $\beta_1 \rightarrow$ pendiente
 tasa de cambio
 peso



¿El mejor modelo $\hat{y}$?

$$\hat{y} = \beta_0 + \beta_1 x$$

pred.

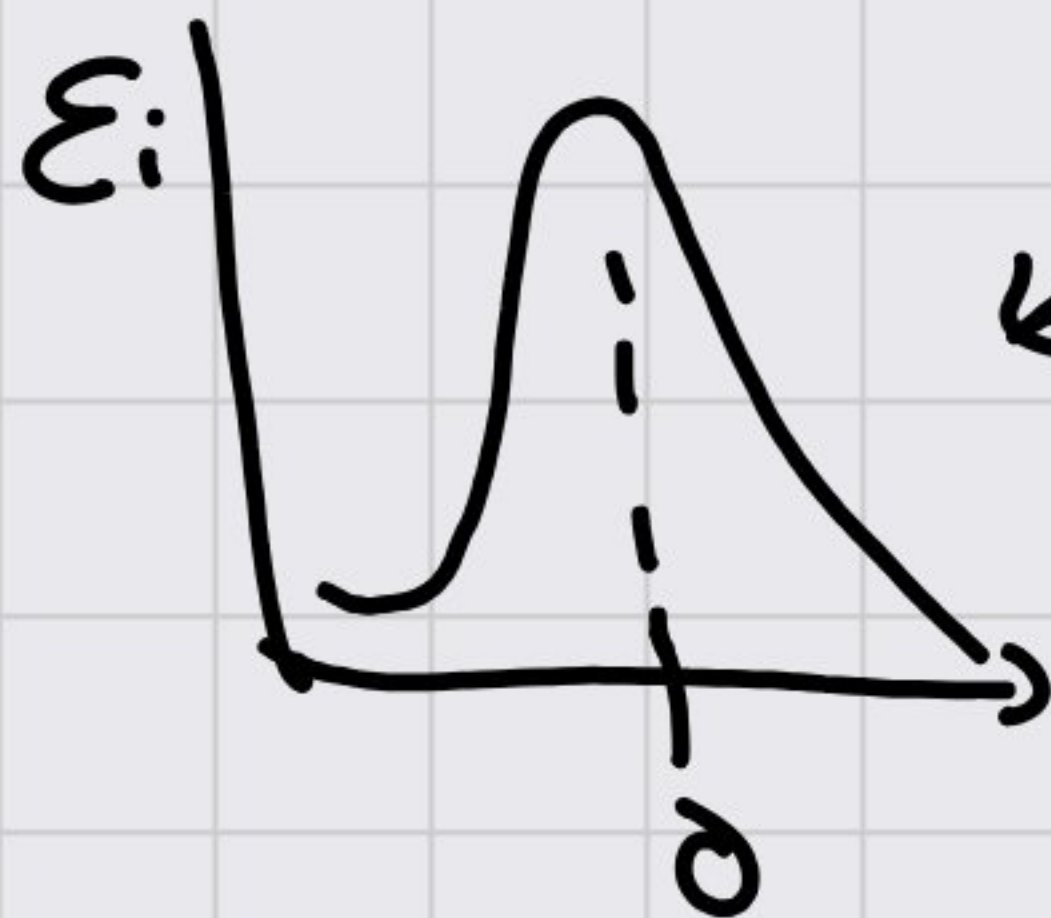
Datos
 x_i, y_i

Función de costo ($\mathcal{L}$)

$$\varepsilon_i = \hat{y}_i - y_i$$

residuo

Pérdida



Mean
Bias Error

$$\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)$$

Mean Squared
Error,

$$\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$$

MSE

Mean Absolute
Error,

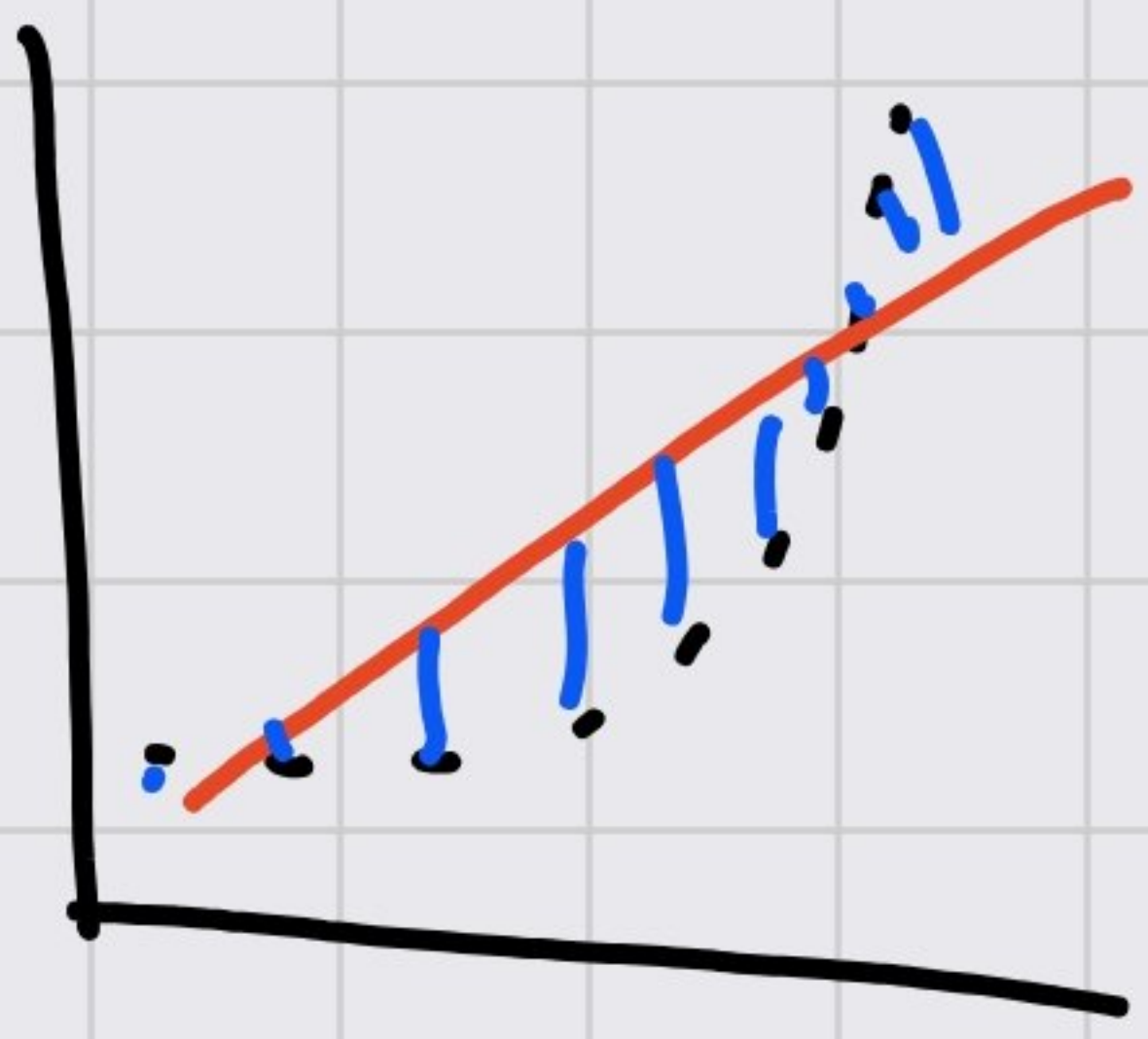
$$\frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i|$$

MAE

Root Mean
Squared Error

$$\sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2}$$

RMSE



Modelo

$$\hat{y} = \beta_0 + \beta_1 x$$

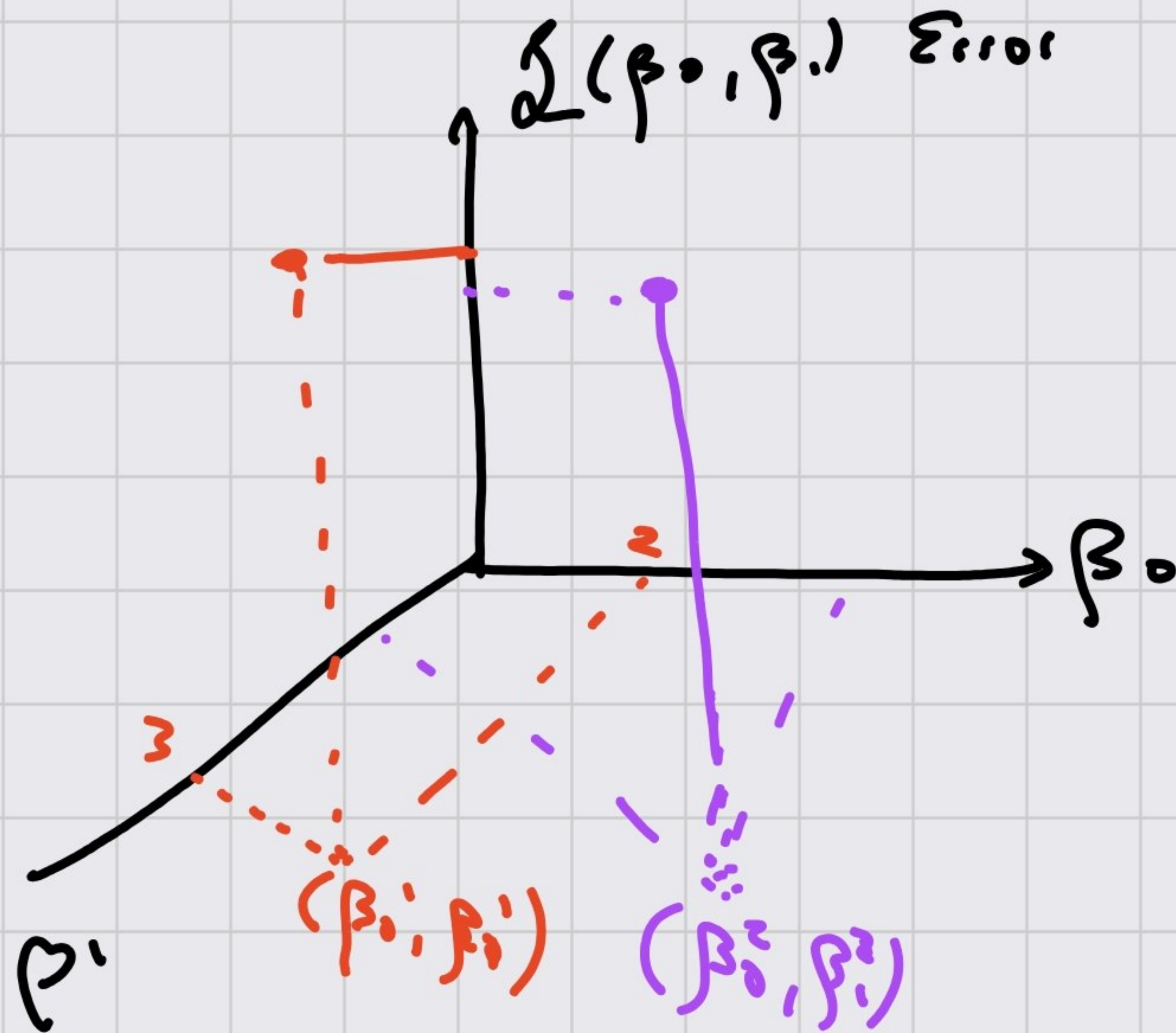
Función de Costo
Error

$$J = \frac{1}{n} \sum (\hat{y}_i - y_i)^2$$

$$J(\beta_0, \beta) = \frac{1}{n} \sum (\underbrace{\beta_0}_{\text{green}} + \underbrace{\beta_1}_{\text{green}} \underbrace{x_i}_{\text{blue}} - \underbrace{y_i}_{\text{blue}})^2$$

$$J(\beta_0, \beta_1)$$

$$\vec{\beta} = (\beta_0, \beta_1) \longrightarrow \in \mathbb{R}^{E_{10,1}}$$



$$\mathcal{L} = \frac{1}{n} \sum_i (\hat{y}_i - y_i)^2 = \frac{1}{n} \sum ((\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n) - y_i)^2$$

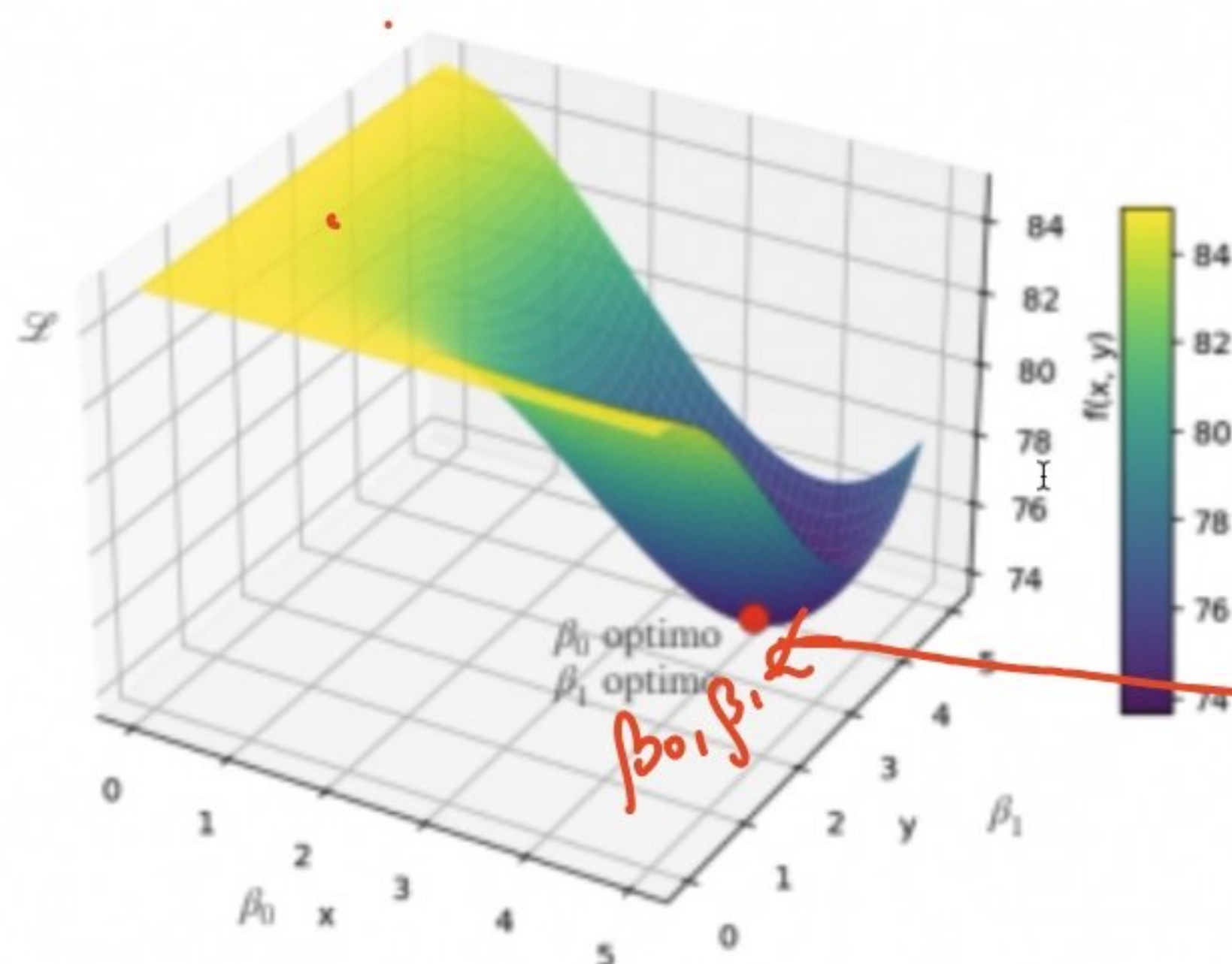
$$\mathcal{L}(\vec{\beta})$$

$$\mathbb{R}^n \rightarrow \mathbb{R}$$

β_0
 β_1

óptimo $\rightarrow \min(\mathcal{L})$

mínimo error



min(L) "mejor vector"

Optimization Error / Cost

$$\min(J), \quad J = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 = \frac{1}{n} \sum_{i=1}^n (\beta_0 + \beta_1 x_i - y_i)^2 \\ = \sum_{i=1}^n u_i^2$$

$$u_i = \beta_0 + \beta_1 x_i - y_i$$

$$\frac{\partial J}{\partial \beta_0} = 0 = \frac{\partial J}{\partial u} \frac{\partial u}{\partial \beta_0} = \left(\sum_{i=1}^n u_i \right) (1)$$

$$\frac{\partial J}{\partial \beta_1} = 0 = \frac{\partial J}{\partial u} \frac{\partial u}{\partial \beta_1} = \left(\sum_{i=1}^n u_i x_i \right)$$

$$\frac{\partial}{\partial \beta_0} L = 0 = \frac{1}{n} \sum u_i = \frac{1}{n} \sum (\beta_0 + \beta_1 x_i - y_i) \quad \dots (1)$$

$$\frac{\partial}{\partial \beta_1} L = 0 = \frac{1}{n} \sum u_i x_i = \frac{1}{n} \sum (\beta_0 + \beta_1 x_i - y_i) x_i \quad \dots (2)$$

Cond. 1 $\frac{1}{n} \sum (\beta_0 + \beta_1 x_i - y_i) = 0 = \frac{1}{n} \sum \beta_0 + \frac{1}{n} \sum \beta_1 x_i - \frac{1}{n} \sum y_i$

$$\Rightarrow \frac{1}{n} \sum y_i = \frac{1}{n} \sum \beta_0 + \frac{1}{n} \sum \beta_1 x_i$$

$$\Rightarrow \boxed{\frac{1}{n} \sum y_i = \beta_0 + \beta_1 \frac{1}{n} \sum x_i} \quad \dots (1)$$

$$\sum_{i=1}^n x_i = x_1 + x_2 + x_3 + \dots + x_n$$

$$\sum_{i=1}^n \beta_0 = \underbrace{\beta_0 + \beta_0 + \dots + \beta_0}_{n \text{ times}} = n\beta_0$$

Cond. 2

$$0 = \frac{1}{n} \sum (\beta_0 + \beta_1 x_i - y_i) x_i = \frac{1}{n} \sum (\beta_0 x_i + \beta_1 x_i^2 - x_i y_i)$$

$$\Rightarrow \frac{1}{n} \sum x_i y_i = \beta_0 \frac{1}{n} \sum x_i + \beta_1 \frac{1}{n} \sum x_i^2$$

... (2)

$$(1) \dots \frac{1}{n} \sum y_i = \beta_0 + \beta_1 \frac{1}{n} \sum x_i$$

Sist. ecuaciones
lineales

$$(2) \dots \frac{1}{n} \sum x_i y_i = \beta_0 \frac{1}{n} \sum x_i + \beta_1 \frac{1}{n} \sum x_i^2$$

Solución: Gauss Jordan, Sustitución,

De la eq. (1)

$$\beta_0 = \underbrace{\frac{1}{n} \sum y_i}_{\bar{y}} - \beta_1 \underbrace{\frac{1}{n} \sum x_i}_{\bar{x}}$$

$$\boxed{\beta_0 = \bar{y} - \beta_1 \bar{x}}$$

Se sustituye en la eq. (2)

$$\dots (2) \quad \frac{1}{n} \sum x_i y_i = \beta_0 \frac{1}{n} \sum x_i + \beta_1 \frac{1}{n} \sum x_i^2$$

$$\text{Subst. } \beta_0 = \bar{y} - \beta_1 \bar{x}$$

$$\begin{aligned} \Rightarrow \frac{1}{n} \sum x_i y_i &= (\bar{y} - \beta_1 \bar{x}) \frac{1}{n} \sum x_i + \beta_1 \frac{1}{n} \sum x_i^2 \\ &= \bar{y} \frac{1}{n} \sum x_i - \beta_1 \bar{x} \frac{1}{n} \sum x_i + \beta_1 \frac{1}{n} \sum x_i^2 \\ &= \bar{x} \bar{y} - \beta_1 \bar{x}^2 + \beta_1 \frac{1}{n} \sum x_i^2 \end{aligned}$$

$$\Rightarrow \underbrace{\left(\frac{1}{n} \sum x_i y_i \right) - \bar{x} \bar{y}}_{\sigma_{xy}} = \beta_1 \underbrace{\left(\left(\frac{1}{n} \sum x_i^2 \right) - \bar{x}^2 \right)}_{\sigma_{x^2}} \quad \therefore \beta_1 = \frac{\sigma_{xy}}{\sigma_{x^2}}$$

Recordamos

$$\sigma_x^2 = \underbrace{\frac{1}{n} \sum (x_i - \bar{x})^2}_{= \frac{1}{n} \sum (x_i^2 - 2x_i\bar{x} + \bar{x}^2)}$$

$$= \frac{1}{n} \sum x_i^2 - 2\bar{x} \frac{1}{n} \sum x_i + \frac{1}{n} \sum \bar{x}^2$$

$$\frac{1}{n} \sum x_i^2 - 2\bar{x}^2 + \bar{x}^2$$

$$= \underbrace{\frac{1}{n} \sum x_i^2 - \bar{x}^2}_{= \sigma_x^2}$$

$$s_{xy} = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

$$= \frac{1}{n} \sum (x_i y_i - x_i \bar{y} - \bar{x} y_i + \bar{x} \bar{y})$$

$$= \frac{1}{n} \sum x_i y_i - \bar{y} \frac{1}{n} \sum x_i - \bar{x} \frac{1}{n} \sum y_i + \frac{1}{n} \sum \bar{x} \bar{y}$$

$$= \underline{\frac{1}{n} \sum x_i y_i - \bar{y} \bar{x}} - \cancel{\bar{x} \bar{y}} + \cancel{\bar{x} \bar{y}}$$

$$= s_{xy}$$

Solución

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

$$\beta_1 = \frac{\sigma_{xy}}{\sigma_x^2}$$

óptimo,
 $\min(\mathcal{L})$

"Mejor recta
posible"

Sol. Analítica

Regresión lineal
simple.